RetailPulse

Autonomous Retail Decision Intelligence

PREDICT. OPTIMIZE. DECIDE. ACT.



Team Name

Team Nexora

Challenge

Challenge 10 – Autonomous Retail Decision Intelligence

Theme

AI for Business

Domain

Retail / CPG

1. The Problem

Retail data is disconnected, forecasts are single numbers, and plans go stale after approval.

Stockouts and overstock at the same time

One store runs out while another holds excess; network-wide options like transfers are missed.

Static rules ignore context

Fixed reorder points do not react to supplier delays, weather, local events or competitor prices.

Forecasts hide uncertainty

A single demand number gives no view of stockout risk or the range of outcomes.

Dashboards report, humans act

Managers must decide and execute manually, and approved plans are not rechecked when conditions change.

2. Proposed Solution

Autonomous Retail Decision Intelligence | Transforming static retail inventory into an intelligent, closed-loop decision system

01. AI Demand Forecasting

Predict store-wise product demand using historical sales patterns, seasonality, and multi-horizon market signals.

02. Dynamic Buffer Stock Optimization [KEY INNOVATION]

Calculate & adjust safety stock from demand swings, supplier lead times and promotions.

03. Smart Inventory Redistribution

Detect surplus stock in over-allocated stores and automatically recommend rebalancing transfers to prevent stockouts.

04. AI-Powered Decision Engine [DECISION CORE]

Evaluate replenishment, transfers, markdowns and promotions for best business actions.

05. What-If Scenario Simulator

Stress-test demand surges, unexpected supplier lead time delays, and promotional shifts before committing capital.

06. Business Impact Analysis

Quantify projected outcomes on stockout rates, excess inventory capital, fulfillment service levels, and net margin.

CENTRAL WORKFLOW & CLOSED-LOOP REASONING

OBSERVE — Data Signals → PREDICT — Demand Waves → OPTIMIZE BUFFER — Dynamic Stock → REASON — Scenario Check → DECIDE — Action Engine → ACT — Execute & Ship → MEASURE — ROI Feedback

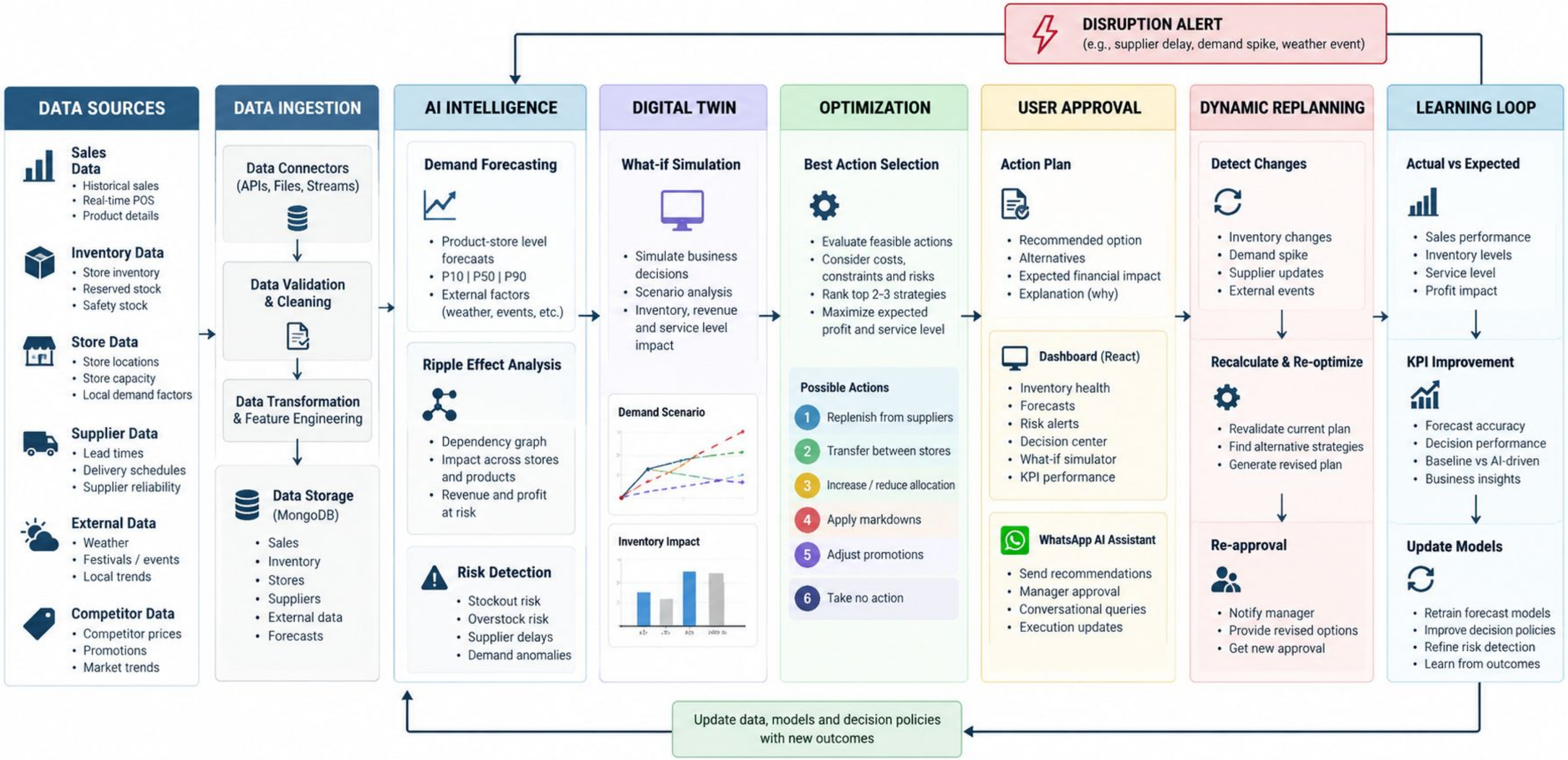
CORE INNOVATION: Proactive Risk Mitigation & Explainable Closed-Loop Execution

RetailPulse proactively identifies store-level inventory stockout and overstock risks, dynamically optimizes safety buffer stocks across regional nodes, and recommends transparent, explainable business decisions to maximize retail margins, operational efficiency, and customer service levels.

3. Tech Stack and Why We Use It

Layer	Technology	Reason
Backend & data	Python, FastAPI, PostgreSQL, Redis	Async APIs and connectors; ACID integrity for orders and approvals; fast live-state cache
Forecasting	HistGradient	Strong on tabular data; quantiles give P10/P50/P90 ranges
Ripple analysis	NetworkX	Dependency graph of suppliers, stores and products for root-cause tracing
Simulation	SimPy, NumPy	Digital Twin of stock flow, delays and demand scenarios (Monte Carlo)
Optimization	Pyomo, HiGHS	Constrained, risk-aware action selection; open-source solver
Agents & language	LangGraph, LLM	Explicit workflow with approval gates; LLM only explains and converses
Interfaces	React, Tailwind CSS, WebSockets, WhatsApp Cloud API	Live dashboard plus the channel managers already use

4. Architecture Diagram



5. Expected Outcomes

- **Fewer stockouts and lost sales**
- **Less overstock and waste**
- **Better stock use across stores**
- **Protected profit margins**
- **Higher customer satisfaction**
- **Faster, explainable decisions**

How we will prove it (benefits are targets to validate, not claimed results)

A. Traditional
Static rules, manual review

B. Prescriptive AI
Forecast, simulate, optimize; manual action

C. Agentic AI
B plus execution and dynamic replanning

Metrics: stockout rate, overstock value, lost sales, net profit, service level, forecast accuracy, decision latency, replanning success rate.